

R Programming Tutorial

#06

Hypothesis Testing 2 Sample

#lectureFIVE-THREE

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#06-1

HYPOTHESIS TESTING 2 SAMPLE IN R

TWO PROPORTION: ONE-TAILED

Hypothesis Testing 1 Sample

Two Proportion

REFER to:
lectureFIVE-THREE #01

EXAMPLE (slide 11-18)

Example

- Below are the sample data and summary statistics:

	Group 1	Group 2
	Subjects Given \$1 Bill	Subjects Given 4 Quarters
Spent the money	$x_1=12$	$x_2=27$
Subjects in group	$n_1=46$	$n_2=43$

$$\hat{p}_1 = \frac{12}{46} \quad \hat{p}_2 = \frac{27}{43}$$

$$\bar{p} = \frac{12+27}{46+43} = 0.438202$$

Hypothesis Testing 1 Sample

Two Proportion

WHAT WE KNOW?

- Two samples:
 - $x_1=12, x_2=27$
 - $n_1=46, n_2=43$
- Significance level: 0.05

WHAT WE WANT TO FIND?

$$H_0 : p_1 = p_2$$

$$H_1 : p_1 < p_2$$

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} \quad \bar{q} = 1 - \bar{p}$$

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\bar{p}\bar{q}}{n_1} + \frac{\bar{p}\bar{q}}{n_2}}}$$

```

1  x1 = 12
2  x2 = 27
3  n1 = 46
4  n2 = 43
5
6  phat1 <- x1/n1
7  phat2 <- x2/n2
8
9  pbar = (x1+x2)/(n1+n2)
10 qbar = 1-pbar
11
12 z = ((phat1-phat2)-
13 0)/sqrt((pbar*qbar/n1)+(pbar*qbar
14 /n2))
15
16 alpha = 0.05
17 z.alpha = qnorm(alpha)
18
19
20 [1] "Critical Value: -1.64485362695147"
21 [1] "Z statistics: -3.48739527429506"
22
  
```

Hypothesis Testing 1 Sample

Two Proportion

REFER to:
lectureFIVE-THREE #01

EXAMPLE (slide 19)


UTM
 UNIVERSITI TEKNOLOGI MALAYSIA

Exercise

www.utm.my

- Time magazine reported the result of a telephone poll of 800 adult Americans. The question posed of the Americans who were surveyed was: "Should the federal tax on cigarettes be raised to pay for health care reform?" The results of the survey were:

Non-smoker	Smoker
$N_1=605$	$N_2=195$
$Y_1=351$ said yes	$Y_2=41$ said yes

- Is there sufficient evidence at the $\alpha = 0.05$ level, say, to conclude that the two populations — smokers and non-smokers — differ significantly with respect to their opinions?

INSPIRING CREATIVE AND INNOVATIVE MINDS

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Hypothesis Testing 1 Sample

Two Proportion

WHAT WE KNOW?

- Two samples:
 - $x_1=351, x_2=41$
 - $n_1=605, n_2=195$
- Significance level: 0.05

WHAT WE WANT TO FIND?

$$H_0 : p_1 = p_2$$

$$H_1 : p_1 \neq p_2$$

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} \quad \bar{q} = 1 - \bar{p}$$

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\bar{p}\bar{q}}{n_1} + \frac{\bar{p}\bar{q}}{n_2}}}$$

```

1  x1 = 351
2  x2 = 195
3  n1 = 605
4  n2 = 195
5
6  phat1 <- x1/n1
7  phat2 <- x2/n2
8
9  pbar = (x1+x2)/(n1+n2)
10 qbar = 1-pbar
11
12 z = ((phat1-phat2)-
13 0)/sqrt((pbar*qbar/n1)+(pbar*qbar
14 /n2))
15
16 alpha = 0.05
17 z.alpha = qnorm(alpha/2)
18
19
20
21 [1] "Critical Value: -1.95996398454005"
22 [1] "Z statistics: -10.9522953986016"
```

#06-2

HYPOTHESIS TESTING 2 SAMPLE IN R

TWO MEANS: KNOWN VARIANCE

Hypothesis Testing 1 Sample

Two Means: Known Variance (σ^2)

REFER to:
lectureFIVE-THREE #02

EXAMPLE (slide 30-32)

Example

A product developer is interested in reducing the drying time of a primer paint. Two formulations of the paint are tested; formulation 1 is the standard chemistry, and formulation 2 has a new drying ingredient that should reduce the drying time. From experience, it is known that the standard deviation of drying time is 8 minutes, and this inherent variability should be unaffected by the addition of the new ingredient. Ten specimens are painted with formulation 1, and another ten specimens are painted with formulation 2; the 20 specimens are painted in random order and normally distributed. The two sample average drying times are $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, respectively. What conclusions can the product developer draw about the effectiveness of the new ingredient, using $\alpha = 0.05$?

Hypothesis Testing 1 Sample

Two Means: Known Variance (σ^2)

WHAT WE KNOW?

- Two samples:
 - $\sigma_1 = \sigma_2 = 8$
 - $n_1=10, n_2=10$
 - $\bar{x}_1 = 121, \bar{x}_2 = 112$

WHAT WE WANT TO FIND?

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 > \mu_2$$

$$Z_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

```

1  xbar1 = 121
2  xbar2 = 112
3
4  n1 = 10
5  n2 = 10
6
7  sigma1 = 8
8  sigma2 = 8
9
10 variance1 = sigma2^2
11 variance2 = sigma2^2
12
13 z = ((xbar1-xbar2-
14 0)/(sqrt((variance1/n1)+(variance
15 2/n2))))
16
17 alpha = 0.05
18 z.alpha = qnorm(alpha,
19 lower.tail=FALSE)
20
21 [1] "Critical Value: 1.64485362695147"
22 [1] "Z statistics: 2.51557647468726"
```

#06-3

HYPOTHESIS TESTING 2 SAMPLE IN R

TWO MEANS: UNKNOWN VARIANCE (EQUAL)

Hypothesis Testing 1 Sample

Two Means: Unknown Variance (Equal σ^2)

REFER to:
lectureFIVE-THREE #02

EXAMPLE (slide 38-42)

■ Example

Two catalysts are being analyzed to determine how they affect the mean yield of a chemical process. Specifically, catalyst 1 is currently in use, but catalyst 2 is acceptable. Since catalyst 2 is cheaper, it should be adopted, providing it does not change the process yield. A test is run in the pilot plant and results in the data shown in Table 5.1. Is there any difference between the mean yields? Use $\alpha = 0.05$, and assume equal variances.

Hypothesis Testing 1 Sample

Two Means: Unknown Variance (Equal σ^2)

WHAT WE KNOW?

- Two samples:
 - Two sets of data
 - Significance: 0.05

WHAT WE WANT TO FIND?

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$$t_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

```

1  xbar1 = 121
2  xbar2 = 112
3
4  n1 = 10
5  n2 = 10
6
7  sigma1 = 8
8  sigma2 = 8
9
10 variance1 = sigma2^2
11 variance2 = sigma2^2
12
13 z = ((xbar1-xbar2-
14 0)/(sqrt((variance1/n1)+(variance
15 2/n2))))
16
17 alpha = 0.05
18 z.alpha = qnorm(alpha,
19 lower.tail=FALSE)
20
21 [1] "Critical Value: -2.1447866879178 2.1447866879178"
22 [1] "T statistics: -0.35359086434619"
```

#06-3

HYPOTHESIS TESTING 2 SAMPLE IN R

TWO MEANS: UNKNOWN VARIANCE (UNEQUAL)

Hypothesis Testing 1 Sample

Two Means: Unknown Variance (Unequal σ^2)

REFER to:

lectureFIVE-THREE #02

EXAMPLE (slide 46-50)

Example

Arsenic concentration in public drinking water supplies is a potential health risk. An article in the *Arizona Republic* (Sunday, May 27, 2001) reported drinking water arsenic concentrations in parts per billion (ppb) for 10 metropolitan Phoenix communities and 10 communities in rural Arizona. The data is given as follows:

Hypothesis Testing 1 Sample

Two Means: Unknown Variance (Unequal σ^2)

WHAT WE KNOW?

- Two samples:
 - Two sets of data
 - $s_1 = 7.63$, $s_2 = 15.3$
 - $n_1 = 10$, $n_2 = 10$
 - $\bar{x}_1 = 12.5$, $\bar{x}_2 = 27.5$
 - Significance: 0.05

WHAT WE WANT TO FIND?

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

$$v = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} \right)^2}{\frac{\left(\frac{S_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2} \right)^2}{n_2 - 1}}$$

$$T_0^* = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

```

1  xbar1 = 12.5
2  xbar2 = 27.5
3
4  s1 = 7.63
5  s2 = 15.3
6
7  n1 = 10
8  n2 = 10
9
10 t0 = (xbar1-xbar2-
11 0)/(sqrt((s1^2/n1)+(s2^2/n2)))
12
13 v =
14 ((s1^2/n1)+(s2^2/n2))^2/(((s1^2
15 /n1)^2)/(n1-
16 1))+(((s2^2/n2)^2)/(n2-1)))
17
18 alpha = 0.05
19 t.alpha = qt(alpha/2, floor(v))
20
21

```

```

[1] "Critical Value: -2.16036865646279"
[1] "T statistics: -2.7744169270369"

```

THE END...